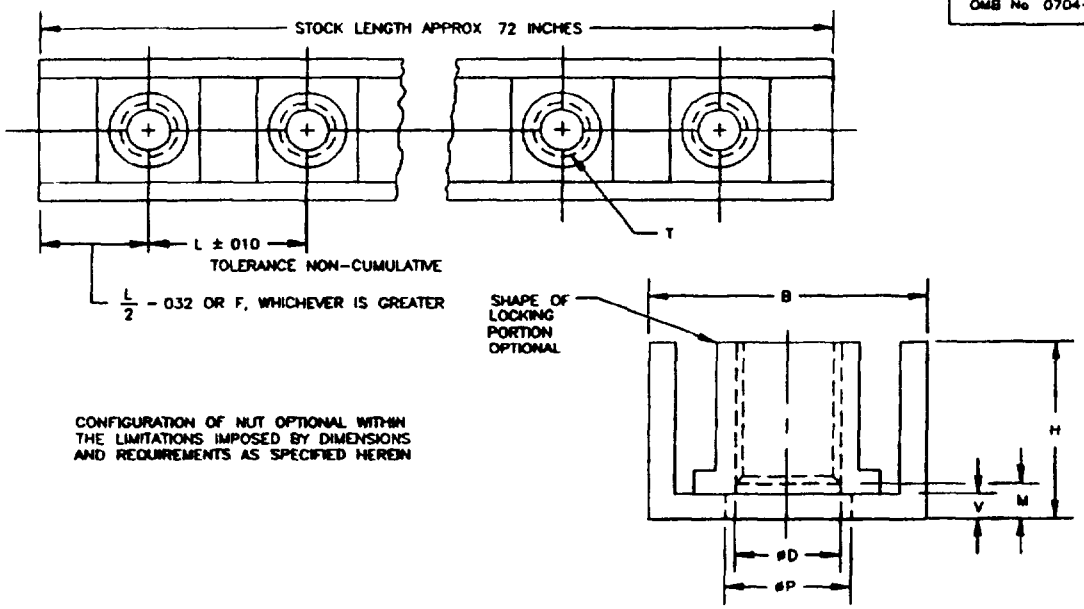
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CONFIGURATION OF NUT OPTIONAL WITHIN THE LIMITATIONS IMPOSED BY DIMENSIONS AND REQUIREMENTS AS SPECIFIED HEREIN

TABLE I DASH NUMBERS AND DIMENSIONS

FIRST DASH NUMBERS		T	B	øD	F	1/ H	L	M	2/ øP		V	AXIAL	WEIGHT	
NON-DRY FILM LUBRICANT	DRY FILM LUBRICANT	THREAD SIZES	MAX	MIN	MIN	MAX	MIN SPACING AVAILABLE	MIN	MAX	MIN	MAX	STRENGTH LBS/MIN	NUT ELEMENTS LBS/100	CHANNEL LBS/INCH
-08	L08	164-32 UNJC-3B	416	168	343	250	625	062	270	184	035	1 720	19	0028
-3	L3	190-32 UNJF-3B	416	194	343	250	625	062	270	210	035	2 460	20	0028
-4	L4	250-28 UNJF-3B	516	254	406	281	750	062	330	270	045	4 580	43	0032
-5	L5	3125-24 UNJF-3B	609	317	469	328	875	062	393	333	045	7 390	64	0053
-6	L6	375-24 UNJF-3B	726	379	562	344	1 000	062	455	395	055	11 450	1 08	0073

1/ "H" MAX APPLIES TO NUT ELEMENT AND CHANNEL. MIN "H" NOT SPECIFIED, LIMITED ONLY BY STRENGTH REQUIREMENTS OF SPECIFICATION
2/ HOLE IN CHANNEL MUST PROVIDE FOR FULL FLOAT OF NUT ELEMENT BUT NEED NOT BE CIRCULAR

TABLE II DASH NUMBERS AND DIMENSIONS

SECOND DASH NUMBER	L NUT ELEMENT SPACING	MAXIMUM NUMBER OF NUT ELEMENT(S)
-5	625	115
-6	750	96
-7	875	82
-8	1 000	72
-9	1 125	64
-10	1 250	57
-11	1 375	52
-12	1 500	48
-13	1 625	44
-14	1 750	41
-15	1 875	38
-16	2 000	36
-18	2 250	32
-20	2 500	28
-24	3 000	24

(H) DENOTES CHANGE(S)

INCH-POUND

PREPARING ACTIVITY DLA-IS
CUSTODIANS ARMY- AV NAVY- AS
AIR FORCE-99 DLA-
REVIEW CR MI
PROJECT NUMBER 5310-1951

MILITARY SPECIFICATION SHEET

TITLE
NUT, SELF-LOCKING GANG CHANNEL, FLOATING, LOW HEIGHT, STEEL, 125 KSI Ftu, 250°F

SPECIFICATION SHEET NUMBER
MS21063 15 APR 94 REV H

SUPERSEDING
MS21063 G 21 NOV 1988

AMSC- N/A FSC 5310

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REQUIREMENTS

1 MATERIAL

NUT ELEMENT - CARBON STEEL COMPOSITIONS 1035 (UNS G10350), 1040 (UNS G10400) AND 1050 (UNS G10500) IN ACCORDANCE WITH ASTM A29, ASTM A827 OR QQ-S-700, 1042 (UNS G10420), IN ACCORDANCE WITH ASTM A29 ALLOY STEEL, GRADES 4130 (UNS G41300) 4340 (UNS G43400) AND 8740 (UNS G87400) IN ACCORDANCE WITH ASTM A29

CHANNEL - ALUMINUM ALLOY 1100 ANNEALED (UNS A91100) IN ACCORDANCE WITH QQ-A-250/1 ALUMINUM ALLOY 2024, TEMPER T4 (UNS A92024) IN ACCORDANCE WITH QQ-A-250/4 ALUMINUM, ALLOY 7075, TEMPER T6 (UNS A97075) IN ACCORDANCE WITH QQ-A-250/12 OR QQ-A-250/13 (ALCLAD)

2 LENGTH. TO BE STOCKED IN 72-INCH LENGTHS ONLY

3 FINISH. NUT ELEMENTS - CADMIUM PLATE IN ACCORDANCE WUTH QQ-P-416 TYPE II CLASS 2 DRY FILM LUBRICATED NUT ELEMENTS IN ACCORDANCE WITH QQ-P-416 TYPE AND CLASS ARE OPTIONAL IF THE NUT ELEMENTS WILL MEET THE SALT SPRAY REQUIREMENTS OF QQ-P-416 TYPE II
CHANNEL - IN ACCORDANCE WITH PROCUREMENT SPECIFICATION

4 LUBRICANT. DRY FILM LUBRICANT ON NUT ELEMENT ONLY APPROVED IN ACCORDANCE WITH MIL-N-25027 NON-DRY FILM LUBRICANTS SHALL BE SOLUBLE IN THE CLEANER SPECIFIED IN MIL-S-6802

5 DIMENSIONING AND TOLERANCING. DIMENSIONING AND TOLERANCING SHALL BE IN ACCORDANCE WITH ANSI Y14 5M

6 HARDNESS. NUT ELEMENTS SHALL HAVE A HARDNESS OF 49HRC MAX

7 THREADS. THREADS BEFORE LUBRICATION IN ACCORDANCE WITH MIL-S-8879

8 SURFACE TEXTURE. SURFACE TEXTURE UNLESS OTHERWISE SPECIFIED SHALL NOT EXCEED 125 MICROINCHES, IN ACCORDANCE WITH ANSI/ASME B46 1

9 FLOAT OF NUT ELEMENT. FLOAT OF NUT ELEMENT PORTION OF ASSEMBLY SHALL NOT BE LESS THAN 030 NOR MORE THAN 040 LONGITUDINALLY AND NOT LESS THAN 010 NOR MORE THAN 030 LATERALLY FROM CENTERED POSITION NUT ELEMENT SHALL BE CAPABLE OF ENGAGEMENT WITH A BOLT IN THE MAXIMUM MISALIGNMENT POSITION MAXIMUM AXIAL FLOAT 020 INCH FOR -08 AND -3 030 FOR LARGER SIZES

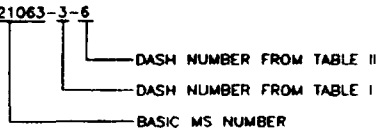
10 ASSEMBLY. THE ASSEMBLY SHALL PROVIDE A BEARING SURFACE FOR THE NUT ELEMENT WITHIN THE HOUSING THE CENTERLINE OF THE CHANNEL SHALL NOT DEVIATE FROM A STRAIGHT LINE BY MORE THAN 015 IN ANY 12 INCHES THE NUT ELEMENT AND BASE PORTION OF THE ASSEMBLY SHALL FORM ONE INTEGRAL UNIT

11 EDGES. BREAK SHARP CORNERS AND REMOVE ALL BURRS

12 CODE. FIRST DASH NUMBER DESIGNATES THREAD SIZE
SECOND DASH NUMBER DESIGNATES NUT ELEMENT SPACING IN EIGHTHS OF AN INCH
THIRD DASH NUMBER INDICATES NUMBER OF NUT ELEMENTS WHEN LESS THAN STOCK LENGTH (APPROX 72 INCHES) IS DESIRED
LETTER "L" BEFORE FIRST DASH NUMBER DESIGNATES DRY FILM LUBRICATED NUT ELEMENTS

13 PART NUMBER. THE PART NUMBER SHALL CONSIST OF THE BASIC MS NUMBER FOLLOWED BY A DASH NUMBER FROM TABLE I AND TABLE II AND A THIRD DASH NUMBER IF SPECIFIED (SEE REQUIREMENT 12)

EXAMPLE MS21063-3-6



MS21063-3-6 INDICATES NUT SELF-LOCKING GANG CHANNEL, FLOATING LOW HEIGHT, STEEL 125 KSI Ftu, 250° F, GANG CHANNEL ASSEMBLY CONSISTING OF NINETY-SIX 190-32 UNJF-3B CADIUM PLATED NUT ELEMENTS SPACED AT 750 INCH

MS21063L4-7-10 INDICATES NUT SELF-LOCKING GANG CHANNEL FLOATING LOW HEIGHT STEEL, 125 KSI Ftu 250° F GANG CHANNEL ASSEMBLY CONSISTING OF TEN 250-28 UNJF-3B CADMIUM PLATED DRY FILM LUBRICATED NUT ELEMENTS SPACED AT 875 INCH

PREPARING ACTIVITY DLA-IS CUSTODIANS ARMY- AV NAVY- AS AIR FORCE-99 DLA-- REVIEW CR, MI PROJECT NUMBER 5310-1951	MILITARY SPECIFICATION SHEET TITLE NUT, SELF-LOCKING GANG CHANNEL, FLOATING, LOW HEIGHT, STEEL, 125 KSI Ftu, 250°F	SPECIFICATION SHEET NUMBER MS21063 15 APR 94 REV H SUPERSEDING MS21063 G 21 NOV 1988 AMSC- N/A FSC 5310
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NOTES

1 ALL DIMENSIONS ARE IN INCHES DIMENSIONS TO BE MET PRIOR TO LUBRICATION

2 IN THE EVENT OF A CONFLICT BETWEEN THE TEXT OF THIS STANDARD AND THE REFERENCES CITED HEREIN THE TEXT OF THIS STANDARD SHALL TAKE PRECEDENCE

3 REFERENCED GOVERNMENT (OR NON-GOVERNMENT) DOCUMENTS OF THE ISSUE LISTED IN THAT ISSUE OF THE DEPARTMENT OF DEFENSE INDEX OF SPECIFICATIONS AND STANDARDS (DODISS) SPECIFIED IN THE SOLICITATION FORM A PART OF THIS STANDARD TO THE EXTENT SPECIFIED HEREIN

4 DESIGN AND USAGE LIMITATIONS THESE NUTS ARE DESIGNED TO DEVELOP THE TENSILE STRENGTHS OF BOLTS AND SCREWS WITH AN ULTIMATE TENSILE STRENGTH OF 125 KSI BASED ON THE CROSS SECTION AREA AT THE BASIC ROOT DIAMETER OF THE THREADS THESE NUTS ARE DESIGNED TO BE USED ON 3A EXTERNAL THREADS THESE NUTS SHALL BE USED IN ACCORDANCE WITH THE LIMITATIONS OF MS33588 ONLY NUTS FOR WHICH THERE ARE QUALIFIED PRODUCTS LISTED ON QPL 25027 SHALL BE USED

INTERCHANGEABILITY RELATIONSHIP

MS21063 NUTS CAN UNIVERSALLY REPLACE NAS688 THRU NAS692 AND NAS1034 THRU NAS1038 NUTS OF LIKE THREAD SIZE LUBRICANT (DRY FILM OR NON-DRY FILM), PLATING NUT ELEMENT SPACING NUMBER OF NUT ELEMENTS AND MATERIAL BUT THESE NAS688 THRU NAS692 AND NAS1034 THRU NAS1038 NUTS CANNOT UNIVERSALLY REPLACE MS21063 NUTS

INTERCHANGEABILITY TABLE

CANCELLED PART NUMBERS 1/		SUBSTITUTIVE PART NUMBERS 1/
NAS1034PX	NAS688PX	MS21063-08
NAS1035PX	NAS689PX	MS21063-3
NAS1036PX	NAS690PX	MS21063-4
NAS1037PX	NAS691PX	MS21063-5
NAS1038PX	NAS692PX	MS21063-6
NAS1034P	NAS688P	MS21063L08
NAS1035P	NAS689P	MS21063L3
NAS1036P	NAS690P	MS21063L4
NAS1037P	NAS691P	MS21063L5
NAS1038P	NAS692P	MS21063L6

1/ NUT ELEMENT SPACING DASH NUMBER IS IDENTICAL FOR BOTH CANCELLED AND CORRESPONDING SUBSTITUTIVE PART NUMBERS

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